

Assignment 1

Information Theory for Data Science

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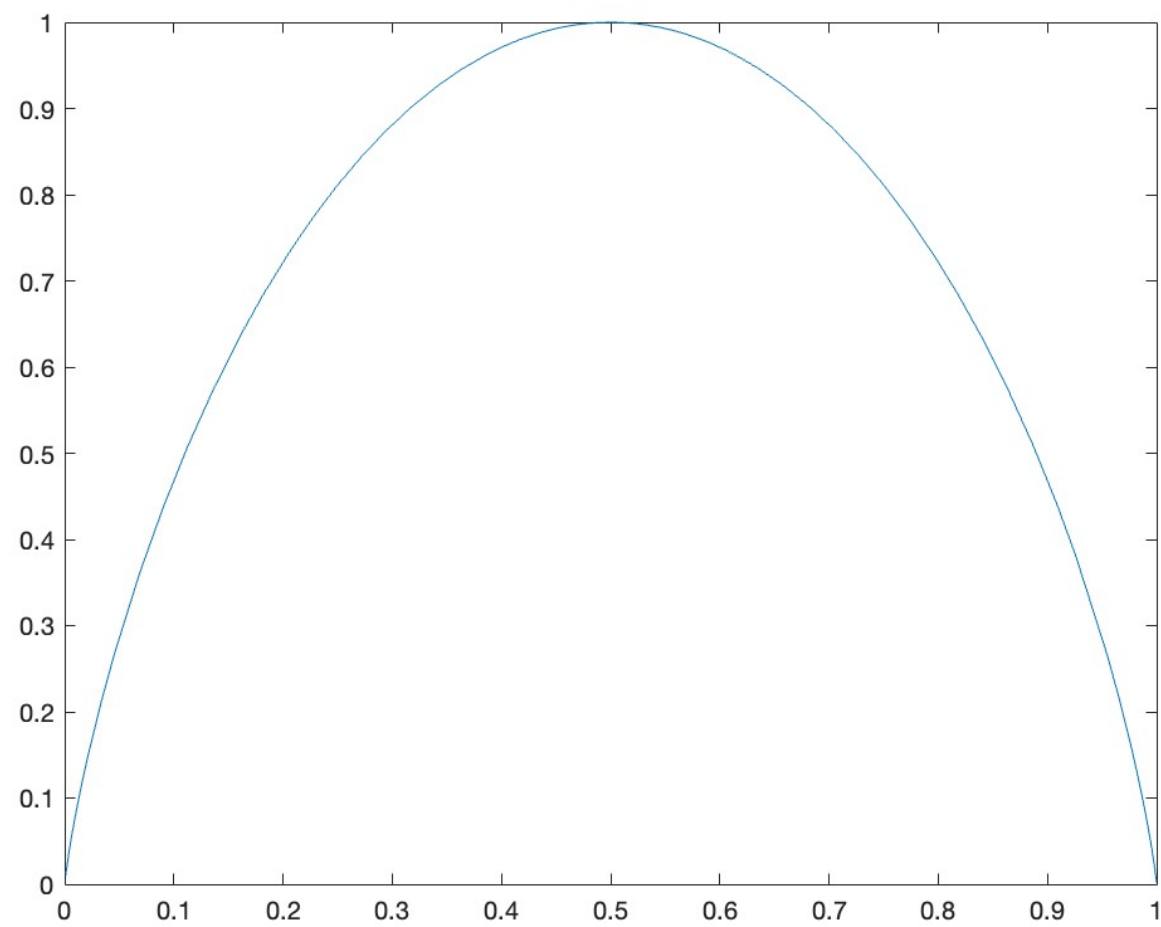
Exercise 1

Entropy of a binary random variable

Exercise 1.A

1. Plot the entropy of a binary random variable
2. Discuss the result

1.



2.

When $P=0$ then the other outcome ($1-P=1$) is definite, so there is no uncertainty and the entropy is 0.

When $P=1/2$ both outcomes are equal so the uncertainty is maximum, and we have the maximum amount of information. Therefore, the Entropy is maximum and equal to 1.

When $P=1$ then the outcome is definite, so there is no uncertainty and the entropy is 0.

Exercise 1.B – Renyi entropy

$$H_{\alpha}(X) = \frac{1}{1-\alpha} \log_2 \left[\sum_i p_i^{\alpha} \right] \quad \alpha \geq 0 \quad \alpha \neq 1$$

3. Prove that the limit for $\alpha \rightarrow 1$ is the Shannon entropy.
4. Repeat point 1 with different values of α (smaller and bigger than 1)
5. Comment the results

3.

$$H_{\alpha}(X) = \frac{1}{1-\alpha} \log_2 \left[\sum_i p_i^{\alpha} \right] \quad \alpha \geq 0, \alpha \neq 1$$

Shannon Entropy: $H(X) = - \sum p(x) \log_2 p(x)$

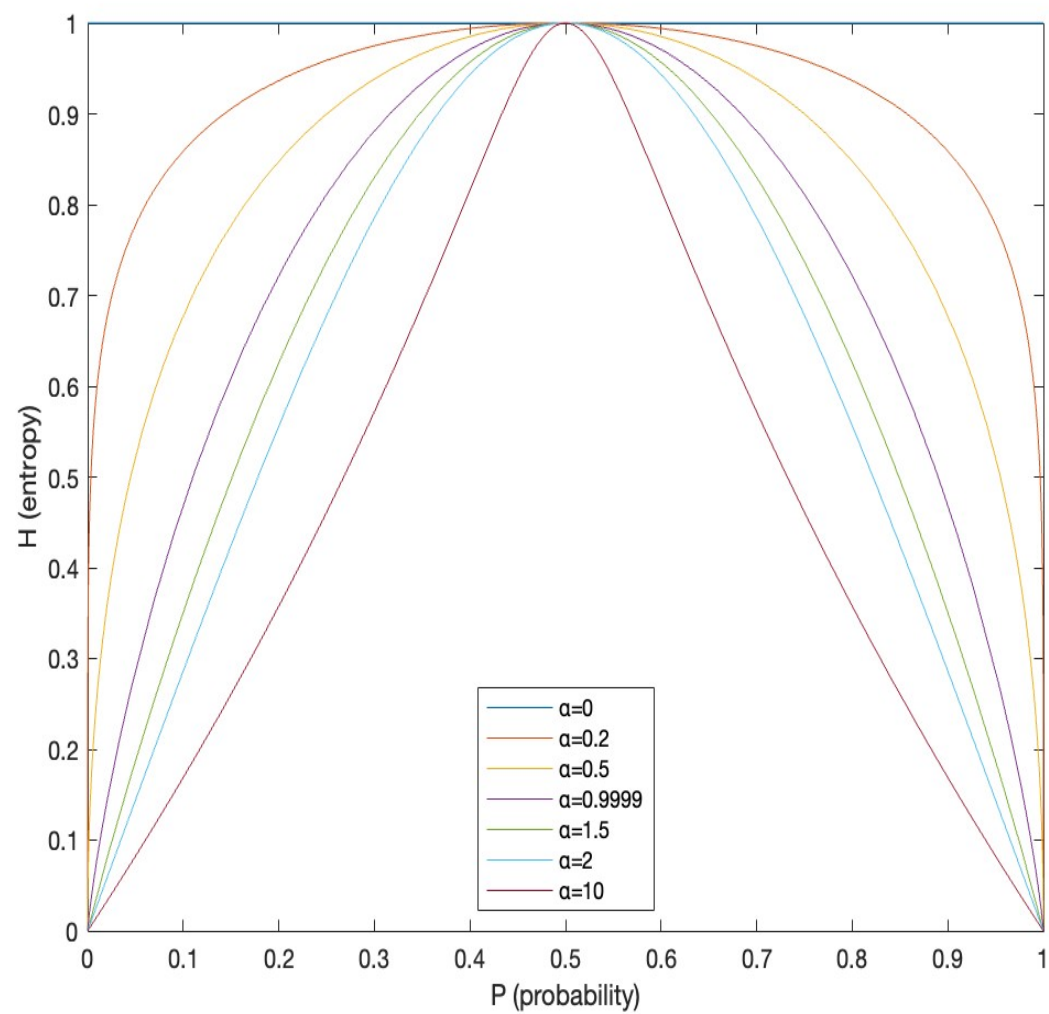
$$\lim_{\alpha \rightarrow 1} H_{\alpha}(X) = \lim_{\alpha \rightarrow 1} \frac{1}{1-\alpha} \log_2 \left[\sum_i p_i^{\alpha} \right]$$

Hopital $\rightarrow$
$$\frac{\frac{d}{d\alpha} \log_2 \left[\sum_i p_i^{\alpha} \right]}{\frac{d}{d\alpha} (1-\alpha)} = \frac{\frac{p_i^{\alpha} \log p_i}{\log 2 \times p_i^{\alpha}}}{-1} = -\log_2 p_i$$

$$\times \sum_{i=1}^m p_i = 1 \quad \rightarrow \quad -\log_2 p_i \times \sum_{i=1}^m p_i = - \sum p(x) \log_2 p(x)$$

Shannon Entropy

4.



5.

When $0 < \alpha < 1$, the entropy is more sensitive to the higher probabilities like 0.95. It means that the entropy gets low if the uncertainty is very low, which happens when the probability is near 0 or 1.

When $\alpha > 1$, the entropy is more sensitive to lower probabilities like 0.4. It means that the entropy gets higher as the uncertainty increases approximately at an equal rate. When α gets higher the graph becomes linear.

Exercise 1.C

- Prove that this averaging operation always increases the entropy $\{p, p, p_3\}$ $p = \frac{p_1 + p_2}{2}$

Hint: use the log inequality

$$\left. \begin{array}{l} X \quad P_1 \quad P_2 \quad P_3 \\ X' \quad P \quad P \quad P_3 \end{array} \right\} \xrightarrow{P = \frac{P_1 + P_2}{2}} H(X) \leq H(X')$$

$$H(X) = P_1 \log_2 \frac{1}{P_1} + P_2 \log_2 \frac{1}{P_2} + P_3 \log_2 \frac{1}{P_3}$$

$$H(X') = 2P \log_2 \frac{1}{P} + P_3 \log_2 \frac{1}{P_3}$$

$$H(X) - H(X') = \sum_{i=1}^n P_i \log_2 \frac{1}{P_i} - \sum_{j=1}^n \log_2 \frac{1}{P_j}$$

$$\begin{aligned} \xrightarrow{\sum P_i = \sum P_j = 1} \sum P_i \log_2 \frac{P_j}{P_i} &\stackrel{(*)}{\leq} \sum P_i \left(\frac{P_j}{P_i} - 1 \right) \frac{1}{\log_2 e} \\ &= \left(\sum P_j - \sum P_i \right) \frac{1}{\log_2 e} = 0 \end{aligned}$$

$$\Rightarrow H(X) - H(X') \leq 0 \Rightarrow H(X) \leq H(X')$$

$$(*) \text{ log inequality: } \log_2 \frac{P_j}{P_i} = \frac{\log_e \frac{P_j}{P_i}}{\log_e 2} \leq \frac{\frac{P_j}{P_i} - 1}{\log_e 2}$$

Exercise 2

Application of the principle of maximum entropy

1. Invent an exercise where you have a random variable X with alphabet Ω_X where each outcome has a given “cost”.
2. Fix the mean value bigger than the arithmetic average of the costs, and apply the principle of maximum entropy to find the probability distribution $P(X)$
3. Plot $P(X)$
4. Repeat with a mean value equal to the arithmetic average and plot the result
5. Repeat with other values of the mean value and plot the results
6. Comment the results

2.1.

$\Omega(x) = \{ \text{"t-shirt"}, \text{"jacket"}, \text{"pants"}, \text{"gloves"}, \text{"shoes"} \}$

$\text{cost}(x) = \{20, 50, 30, 15, 25\}$

arithmetic average = $\frac{20+50+30+15+25}{5} = 28$

2.2. $\mu = 30 > 28$

$$\Delta = f + \lambda_0 g + \lambda_1 h$$

$$= -\sum_i p_i \log_2 p_i + \lambda_0 \sum_i p_i - 1 + \lambda_1 \sum_i \text{cost}(i) p_i - \mu$$

$$\frac{\partial \Delta}{\partial p_i} = 0 \Rightarrow -\log_2 p_i - \cancel{p_i} \log_2 e \frac{1}{\cancel{p_i}} + \lambda_0 + \text{cost}(i) \lambda_1 = 0 \Rightarrow p_i = 2^{\frac{\lambda_0 - \log_2 e}{2}} \cdot \frac{\lambda_1 \text{cost}(i)}{2} = \alpha \beta^{\text{cost}(i)}$$

$$\frac{\partial \Delta}{\partial \lambda_0} = 0 \Rightarrow \sum_i p_i - 1 = 0 \xrightarrow{\text{I}} \alpha \beta^{20} + \alpha \beta^{50} + \alpha \beta^{30} + \alpha \beta^{15} + \alpha \beta^{25} = 1 \Rightarrow \alpha = \frac{1}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}}$$

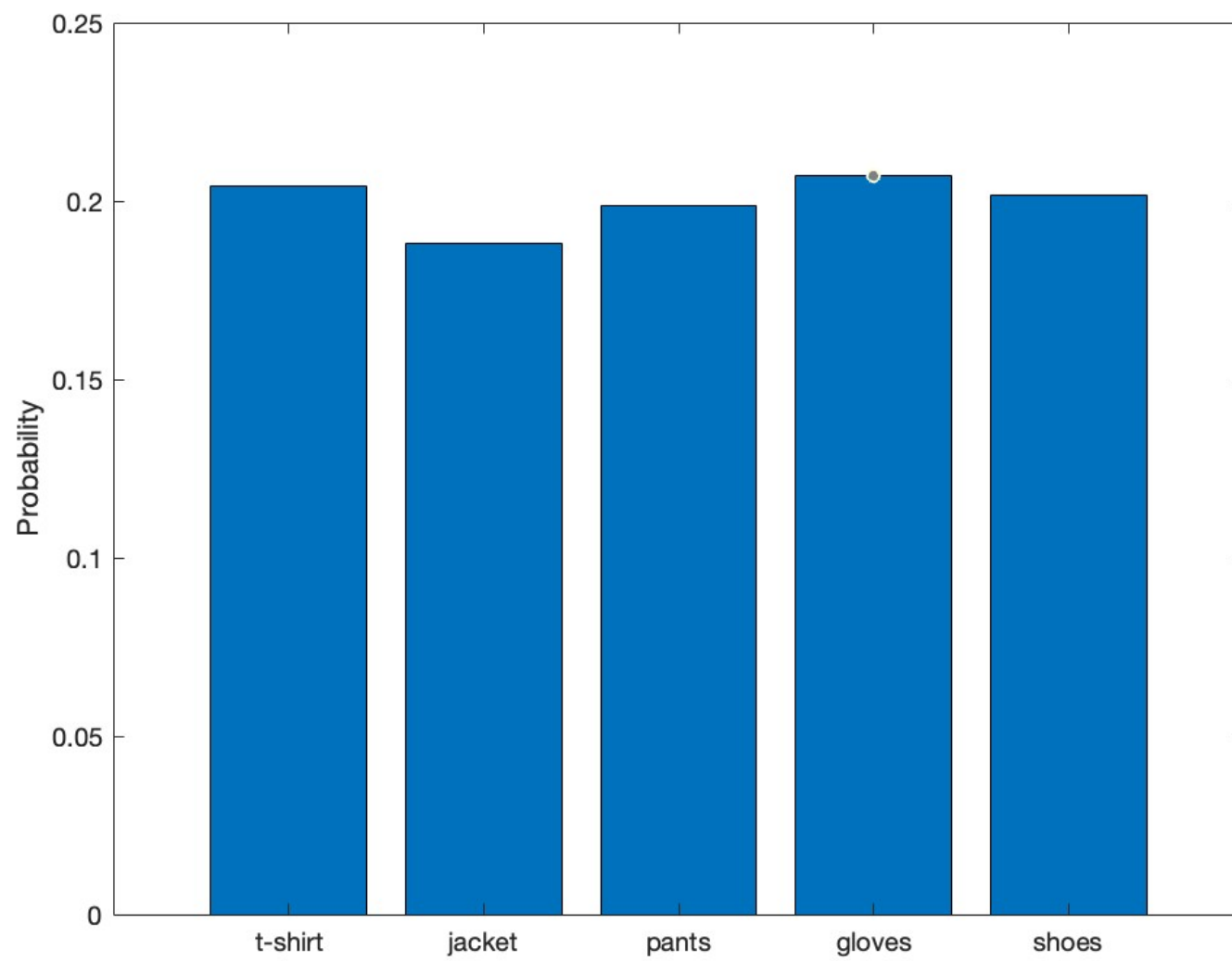
$$p_i = \alpha \beta^i = \frac{\beta^i}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}} \quad \text{II}$$

$$\frac{\partial \Delta}{\partial \lambda_1} = 0 \Rightarrow \sum_i \text{cost}(i) p_i - \mu = 0 \xrightarrow{\text{II}} \sum_{i=1}^5 \frac{\text{cost}(i) \beta^i}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}} = 30$$

$$20\beta + 50\beta^2 + 30\beta^3 + 15\beta^4 + 25\beta^5 = 30(\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25})$$

MATLAB $\Rightarrow \beta \approx 0.9972 \xrightarrow{\text{cost}(i)} \alpha = 0.2160 \Rightarrow p_i = 0.2160 \times 0.9972^{\text{cost}(i)}$

2.3.



2.4. $\mu = 28$

$$\Delta = f + \lambda_0 g + \lambda_1 h$$

$$= -\sum_i p_i \log_2 p_i + \lambda_0 \sum_i p_i - 1 + \lambda_1 \sum_i \text{cost}(i) p_i - \mu$$

$$\frac{\partial \Delta}{\partial p_i} = 0 \rightarrow -\log_2 p_i - \cancel{p_i} \log_2 e \frac{1}{\cancel{p_i}} + \lambda_0 + \text{cost}(i) \lambda_1 = 0 \rightarrow p_i = 2^{\lambda_0 - \log_2 e} \frac{\lambda_1 \text{cost}(i)}{2} = \alpha \beta^{\text{cost}(i)} \quad \textcircled{\text{I}}$$

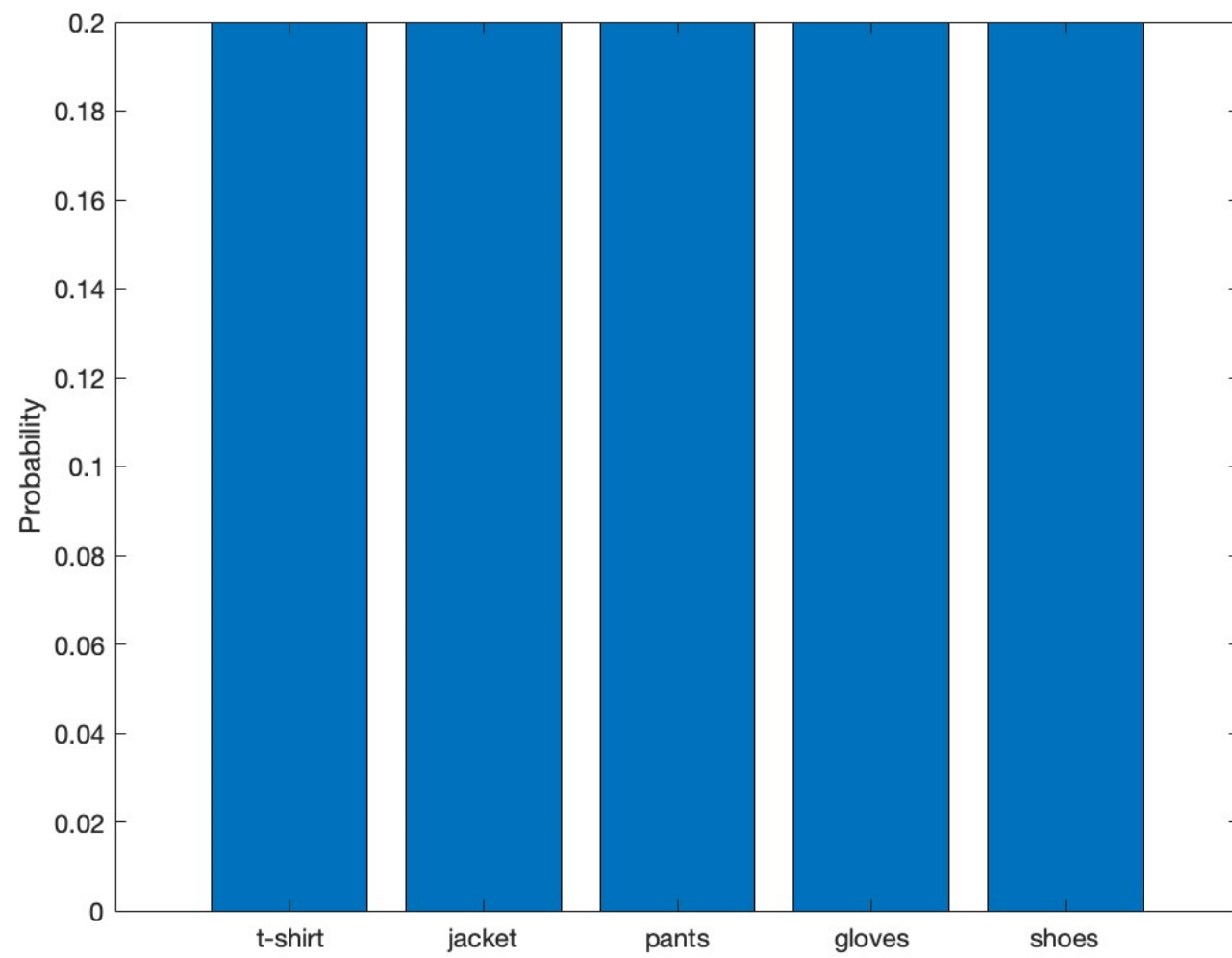
$$\frac{\partial \Delta}{\partial \lambda_0} = 0 \rightarrow \sum_i p_i - 1 = 0 \xrightarrow{\text{I}} \alpha \beta^{20} + \alpha \beta^{50} + \alpha \beta^{30} + \alpha \beta^{15} + \alpha \beta^{25} = 1 \rightarrow \alpha = \frac{1}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}}$$

$$p_i = \alpha \beta^i = \frac{\beta^i}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}} \quad \textcircled{\text{II}}$$

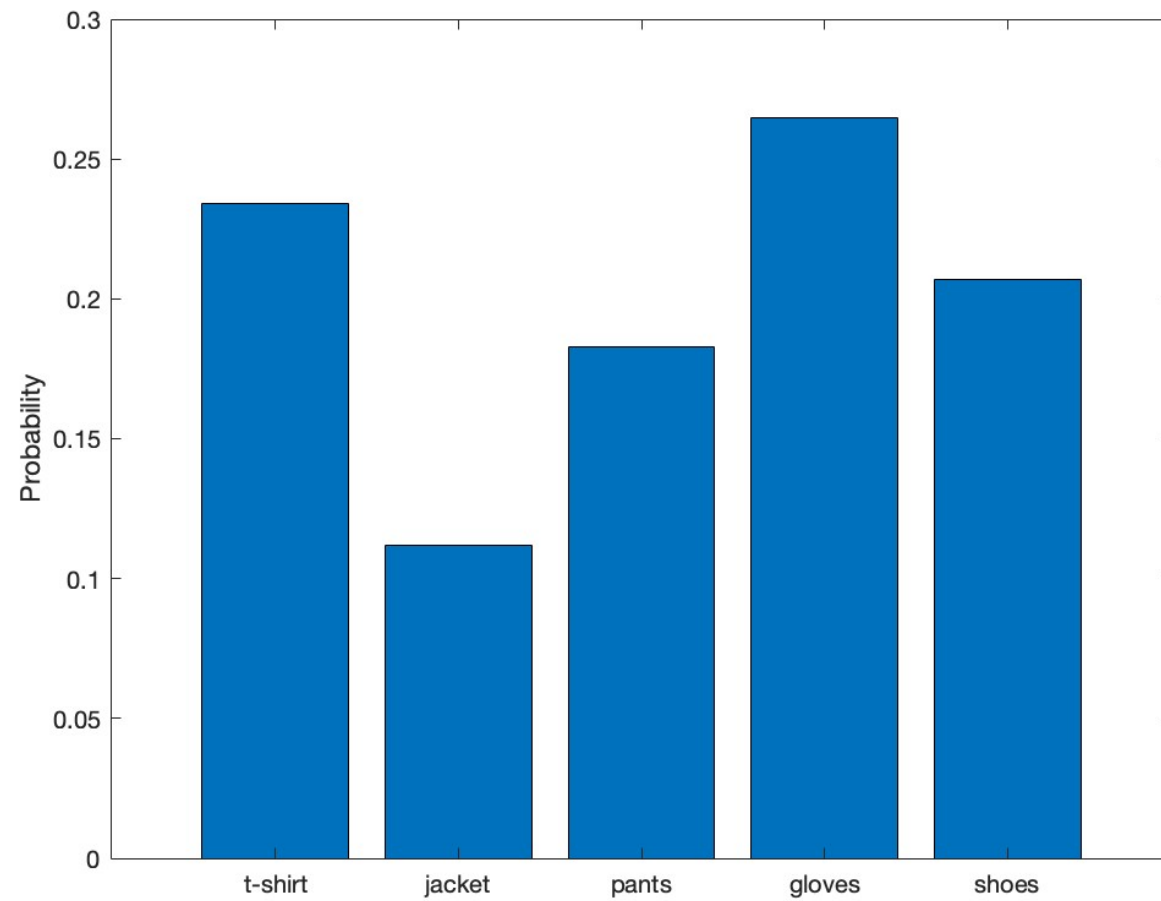
$$\frac{\partial \Delta}{\partial \lambda_1} = 0 \rightarrow \sum_i \text{cost}(i) p_i - \mu = 0 \xrightarrow{\text{II}} \sum_{i=1}^5 \frac{\text{cost}(i) \beta^i}{\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25}} = 28$$

$$20\beta + 50\beta^2 + 30\beta^3 + 15\beta^4 + 25\beta^5 = 28(\beta^{20} + \beta^{50} + \beta^{30} + \beta^{15} + \beta^{25})$$

$$\xrightarrow{\text{MATLAB}} \beta \approx 1 \rightarrow \alpha = 0.2 \rightarrow p_i = 0.2$$



2.5. $\mu = 50$



2.6.

When the mean value is bigger than the arithmetic average of the costs, outcomes with lower costs have a higher probability. We can see this in the figures below which shows the probabilities plot for $\mu=30$ and $\mu=50$ consecutively. As the mean value increases, the probability of higher cost outcomes will decrease.

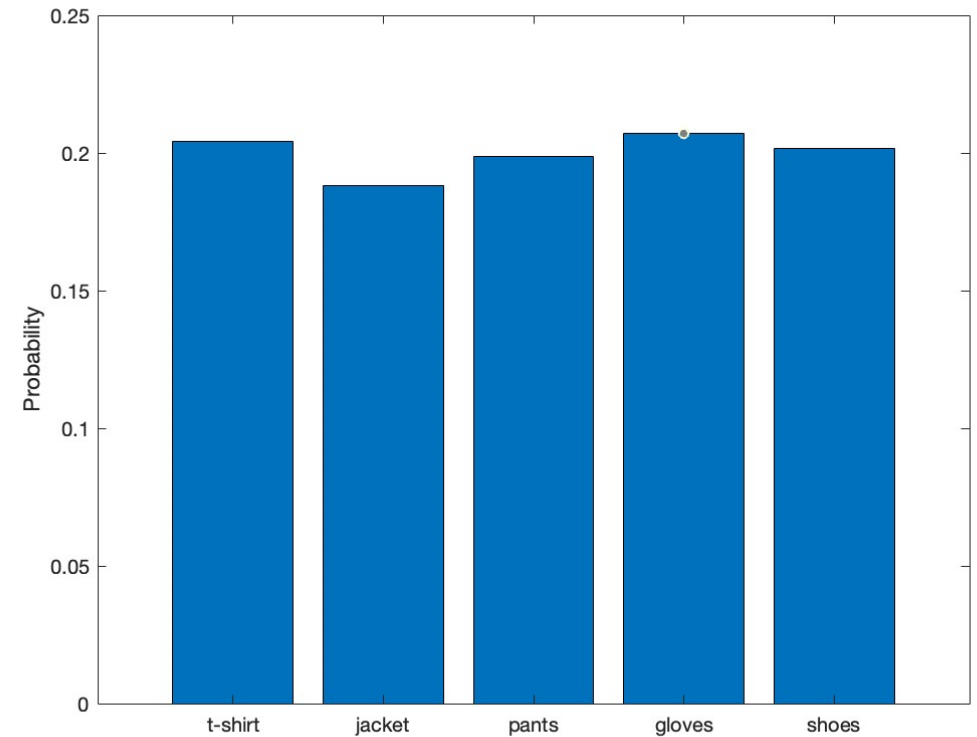


Figure 1- Probability dist. for $\mu=30$

When the mean value is equal to the arithmetic average of the costs, all outcomes have the same probability because beta is 1 and $1^{\text{cost}(i)} = 1$. The plot is shown in the figure below.

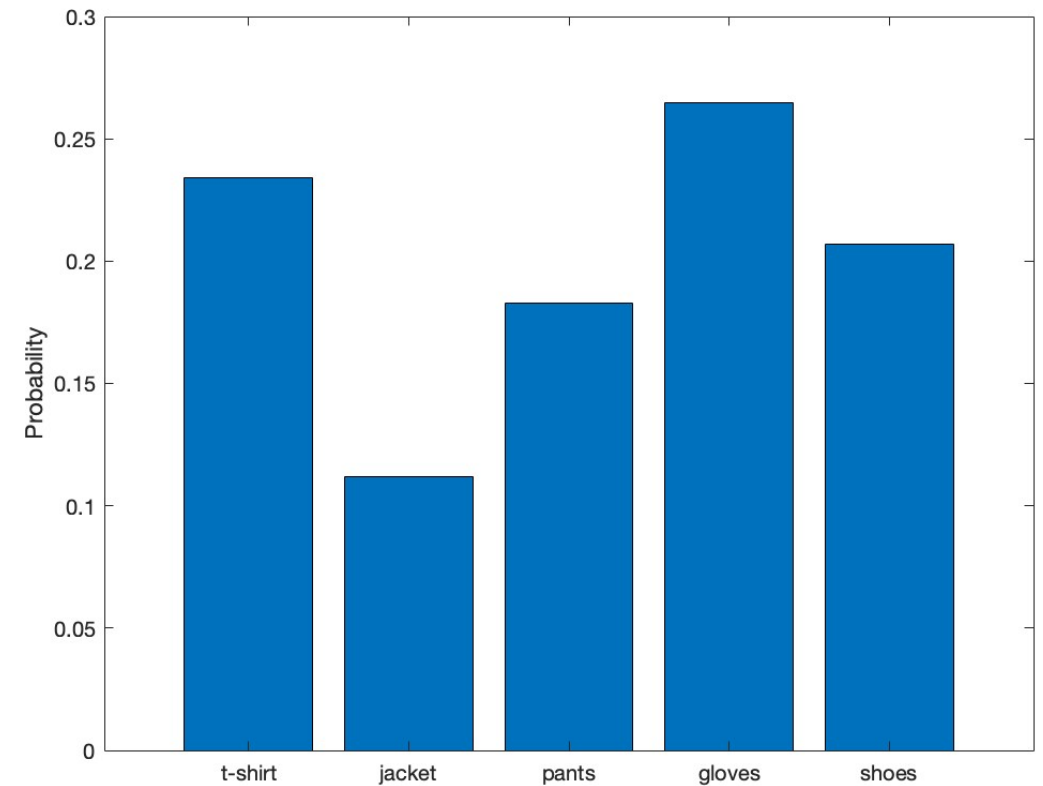


Figure 2- Probability dist. for $\mu=50$

Also, when the mean value is less than the arithmetic average of the costs, outcomes with higher costs have higher probabilities. As the mean value decrease, the probability of outcomes with higher costs increase.

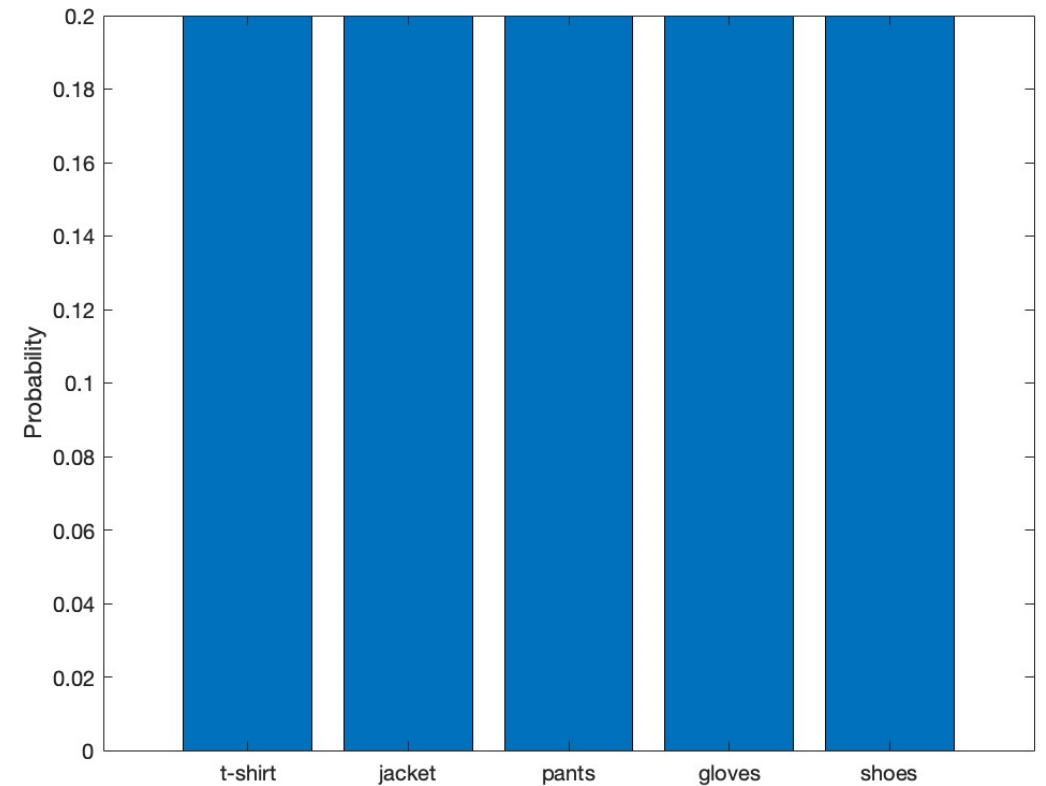


Figure 3- Probability dist. for $\mu=28$

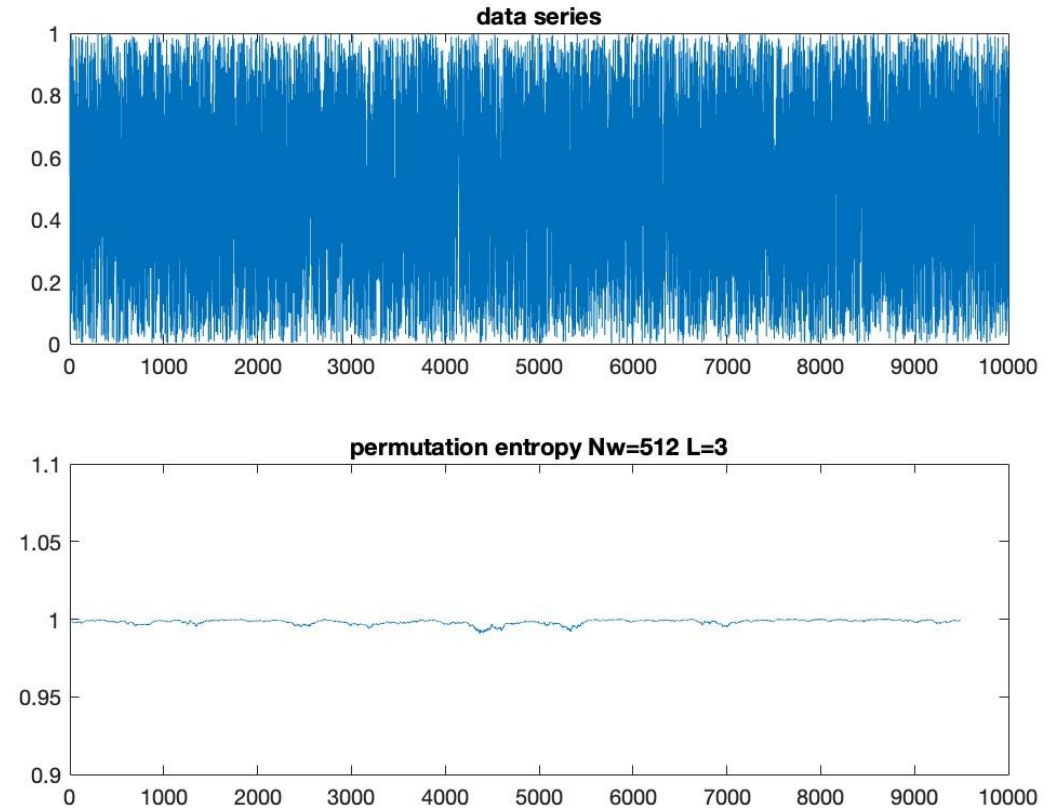
Exercise 3

Permutation Entropy for Time Series
Anomaly Detection

Write a program that:

- Generates a data series made by 10,000 random symbols.
- Computes the permutation entropy with a sliding window of $N_w = 512$ symbols. (For example, choose an order $m = 3$.)
- Plot the data and the entropy.

Here is the plot of data and the entropy:
(We divided the entropy by $\log_2(3!)$ which is the maximum possible value of $\bar{H}$, so the plots will show the value of $\bar{H}$ which is always between zero and one.)



- Insert a pattern between 3000 and 4000 made by correlated data (with about the same mean value and variance of the original random data).
- Describe how you generated the correlated data
- Compute the permutation entropy with the same N_w and m used before.
- Plot the data and the entropy.
- Comment the results

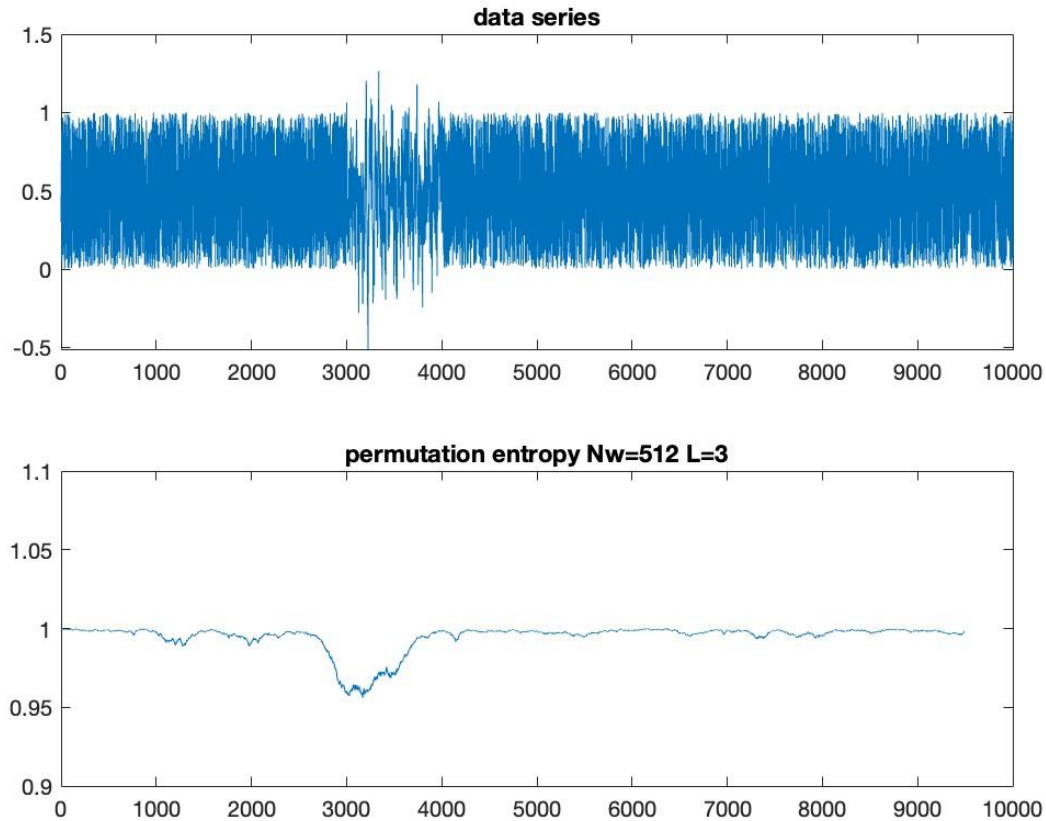
For the second part we generated the correlated data as described below:

First a random array with length of 1000 was produced then to add the correlation, we iterated from the first of the array and then changed the values of array[i] by using the formula below:

$$\text{array}(i) = \text{correlation_coefficient} * \text{array}(i-1) + (1 - \text{correlation_coefficient}) * \text{array}(i)$$

and then we set the mean and variance of the generated array equal to the %95 of the mean and variance of the original data.

Here is the plot of permutation entropy of the data with inserted pattern:



As we can see, when we have correlation in the data even if we have the same mean value and variance, the entropy will decrease because we have less uncertainty as this correlated data brings some predictability and structure to the data.

Exercise 4

Kullback-Leibler distance from
empirical distribution

Consider the observed data

$$x_i = [0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9 \quad 10]$$

$$\text{NUM}(x_i) = [1 \quad 3 \quad 2 \quad 8 \quad 22 \quad 45 \quad 44 \quad 42 \quad 24 \quad 8 \quad 3]$$

1. Compare with uniform pmf
2. Write in the title the KL divergence value

Compare with binomial pmf with $0 < p < 1$ (step=0.001)

3. Identify the pmf at minimum KL divergence
4. Plot the pmf
5. In the title write the value of p and KL divergence

1.

The difference between objects of observed probabilities and uniform probabilities is

```
{0.0049504950495049506,  
 0.009900990099009901,  
 0.01485148514851485,  
 0.039603960396039604,  
 0.10891089108910891,  
 0.1188118811881188,  
 0.2079207920792079,  
 0.21782178217821782,  
 0.22277227722772278}
```

2.

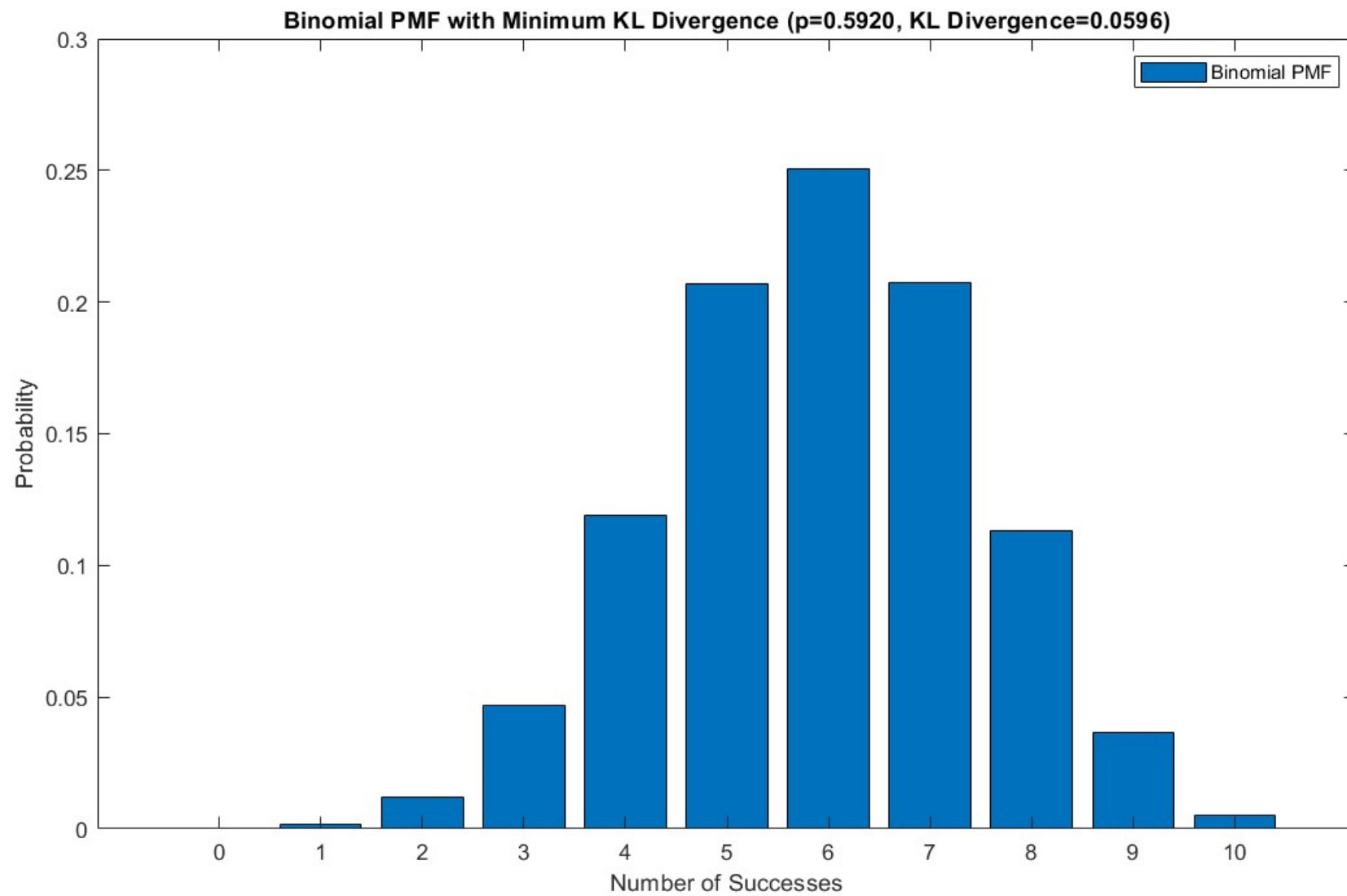
KL divergence value is 0.66004

so the two distributions (observed data probability and uniform probability) are not similar or close to each other.

3.

Among the binomial pmf values, the minimum pmf (minimum Kullback-Leibler Divergence) is 0.05956 and its p-value is 0.592 which is the best p

4,5.



Exercise 5

Information Gain and Tree Classifiers

Given this Training Set

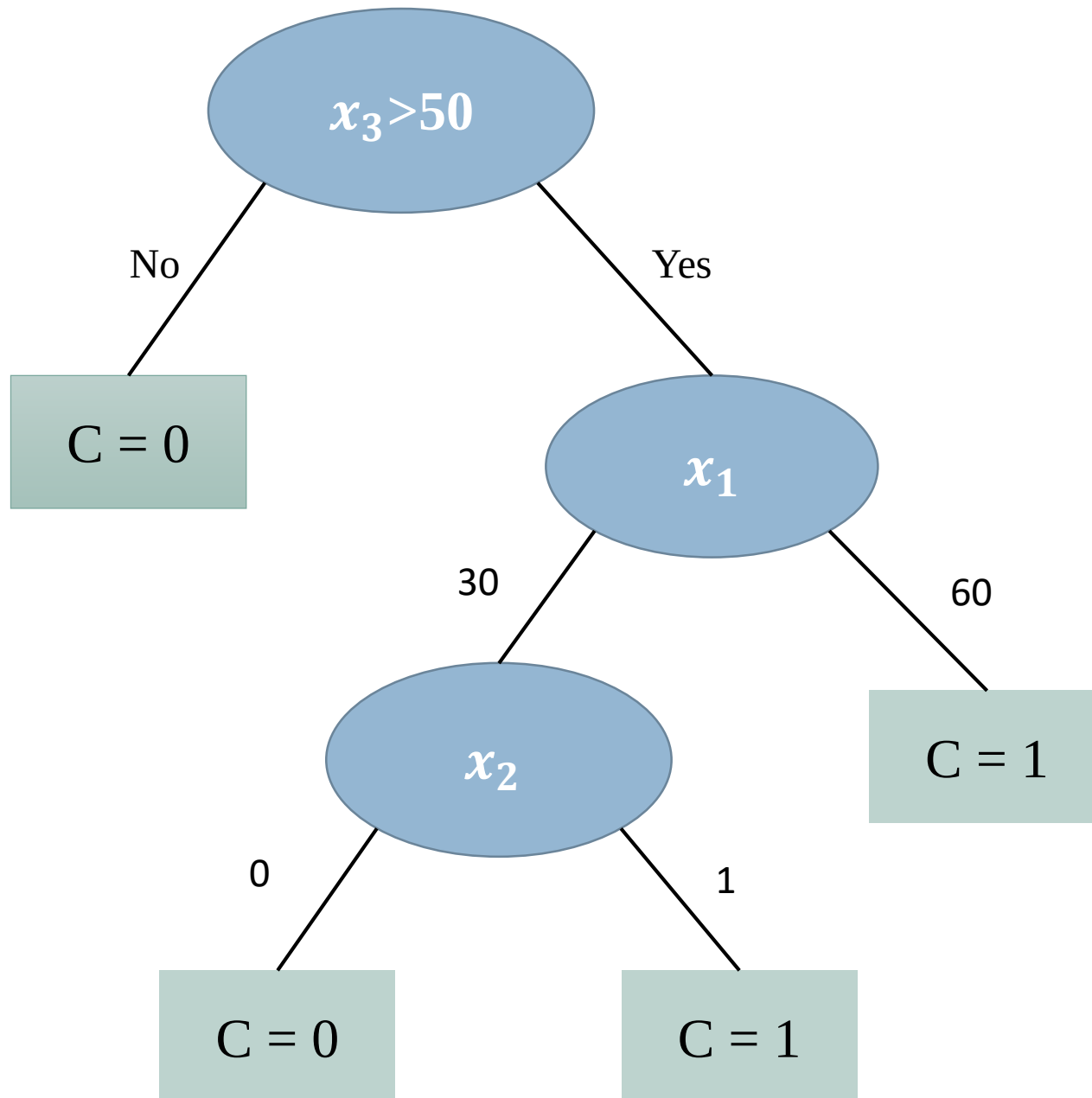
	x_1	x_2	x_3	c
v_1	30	0	10	0
v_2	30	0	70	0
v_3	30	1	20	0
v_4	30	1	80	1
v_5	60	0	40	0
v_6	60	0	60	1
v_7	60	1	50	0
v_8	60	1	60	1

Given this Training Set

Write a program that:

- Builds a tree classifier based on the information gain ratio according to the C4.5 rules for this Training Set (you do not need to write a program able to process any Training set, it is enough if it works for this TS with these three features).
- Accepts a vector $x = (x_1, x_2, x_3)$ as input (from keyboard or file or graphical interface) and outputs the corresponding class.

In the report, plot the tree classifier and present at least 4 examples of class computation.



$$v_1=(30,0,10)$$

$$x_3 > 50 : \text{No} \rightarrow C = 0$$

$$v_2=(30,0,70)$$

$$x_3 > 50 : \text{Yes} \ \& \ x_1 = 30 \ \& \ x_2 = 0 \rightarrow C = 0$$

$$v_3=(30,1,80)$$

$$x_3 > 50 : \text{Yes} \ \& \ x_1 = 30 \ \& \ x_2 = 1 \rightarrow C = 1$$

$$v_4=(60,0,60)$$

$$x_3 > 50 : \text{Yes} \ \& \ x_1 = 60 \rightarrow C = 1$$